

Name: _____

Date: _____

Class: _____

MyDesign Portfolio

Element C Initial Template

How to use this template

Teachers: You may customize this document before providing it to students.

Students: The template below is one way to present your design requirements, but the specific assignment given by your instructor may require a different structure or dictate different details. If you make a copy of this template to edit for your own design requirements, be sure to delete the instructions. Make sure to list your design requirements in ranked order, with design requirement #1 having the highest priority. If possible, list 5 requirements that are labeled as HIGH priority and 1-5 additional requirements that are labeled as LOW priority.

Element C: Presentation and Justification of Solution Design Requirements

Fire extinguisher system-HIGH PRIORITY

The device must be able to put out a small fire inside a box within about 5 seconds using sand or powder. This connects to my research because I learned that sand and fire-extinguishing powder can smother a fire by blocking oxygen, which stops the flames. This is recommended by fire safety organizations like the NFPA. I shared this idea with my teacher, and they said I should include a measurable time, so I added that the fire should be put out within 5 seconds.

Research Stakeholder: use sand/powder for smothering fires.

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Ensuring Complete Fire Coverage to Remove Oxygen-HIGH PRIORITY

The design must fully cover the fire with sand or powder so the flame loses oxygen and stops burning. My research showed that fires need oxygen, heat, and fuel, which is called the fire triangle. If one part is removed, the fire will go out. My teacher agreed with this requirement and suggested making sure the container holds enough sand to cover the entire flame.

Research Stakeholder: Fire-safety sources emphasize smothering as an effective method

Manual Release Mechanism for Fast and Simple Fire Suppression

The system must have a manual way to release the sand or powder quickly. This requirement comes from my research because most fire safety tools like fire extinguishers or sand buckets work by letting someone activate them quickly before the fire spreads. My teacher said the release method should be simple and easy to use, so I planned a simple opening or trigger.

Research Stakeholder: simple and easy to use.

Fire Containment through the box-HIGH PRIORITY

The design must keep the fire contained inside the box so it cannot spread. From my research, fires can grow quickly if they are not contained, so keeping the fire inside a controlled space is important. My stakeholder suggested making sure the box is made from a material that won't easily burn.

Friend: Suggested using a material that won't burn, ensuring safe containment.

Automatic detection systems-HIGH PRIORITY

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This design has to be able to detect the fire on itself and alert everyone that is near the detection system that there is a fire.

This design has connected to my prior research because all the prior attempts I researched on Fire suppression systems, they all had an Automatic detection system and none of them failed and when their system didn't have an Automatic detection system, they failed.

My friend suggested that integrating an automatic detection system would significantly improve the effectiveness of our group's fire suppression design.

Friend: Recommended adding automatic detection to make the system more effective.

A simple fire safety tool

The design should be cheap, meaning the prototype costs less than \$15 in materials. Easy means it can be built in under 45 minutes using scissors, tape/glue, a ruler, and a simple battery powered alarm system. My teacher said keeping the design simple would make it easier to build and test for the project.

Research stakeholder: Said keeping the design simple makes it easier to build and test.

Checking over work-HIGH PRIORITY

This design is putting the details in the fire suppression system and making sure it works properly.

My prior research shows that people who make sure their fire suppression system works before they actually use it have a higher chance of their system not failing.

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My teacher said that it's always great to go over yours before you turn it in or make sure your project works before you officially test it.

Research Stakeholder: Fire-safety sources emphasize testing reliability.